JASPERGOLD FORMAL VRF

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githubUrl: https://github.com/vasanthasricharan/JasperGold-Formal-VRF

Abstract

This project focuses on applying formal verification techniques during RTL verification using Cadence JasperGold. The design under verification is an FSM-based secure arithmetic processing module implemented in Verilog HDL. Formal verification is used to mathematically prove correctness by exploring all reachable states and input combinations, unlike traditional simulation which validates only limited test scenarios. The project verifies FSM transitions, reset behavior, protocol compliance, security handling, deadlock-free operation, unreachable states, and overflow conditions.

Problem Statement

Simulation-based verification using Universal Verification Methodology tests only a limited set of input scenarios, so some corner-case bugs or illegal states may remain undetected.

Project Objective

The objective of this project is to demonstrate the advantages of formal verification over traditional simulation-based verification using Cadence JasperGold. The project aims to prove design correctness mathematically by exhaustively exploring all possible states and conditions within the RTL design.

Why Formal Verification?

- Explores all reachable RTL states.
- Detects hidden corner-case bugs.
- Identifies unreachable and illegal FSM states.
- Verifies deadlock-free operation.
- Improves verification coverage beyond simulation.
- Provides mathematical proof of correctness.

Design Overview

The design is an FSM-based secure arithmetic module implemented in Verilog HDL. It performs arithmetic and logical operations based on the opcode input and includes security and power control logic.

Inputs and Outputs

Inputs	Outputs
clk	grant
rst	result[4:0]
req	valid
data_a[3:0]	error_flag
data_b[3:0]	
opcode[1:0]	

secure_access	
power_enable	

FSM State Description

State	Encoding	Description
IDLE	2'b00	Waits for request signal.
PROCESS	2'b01	Performs operation based on opcode.
DONE	2'b10	Operation completed successfully.
ERROR	2'b11	Indicates security or power failure.

FSM State Diagram

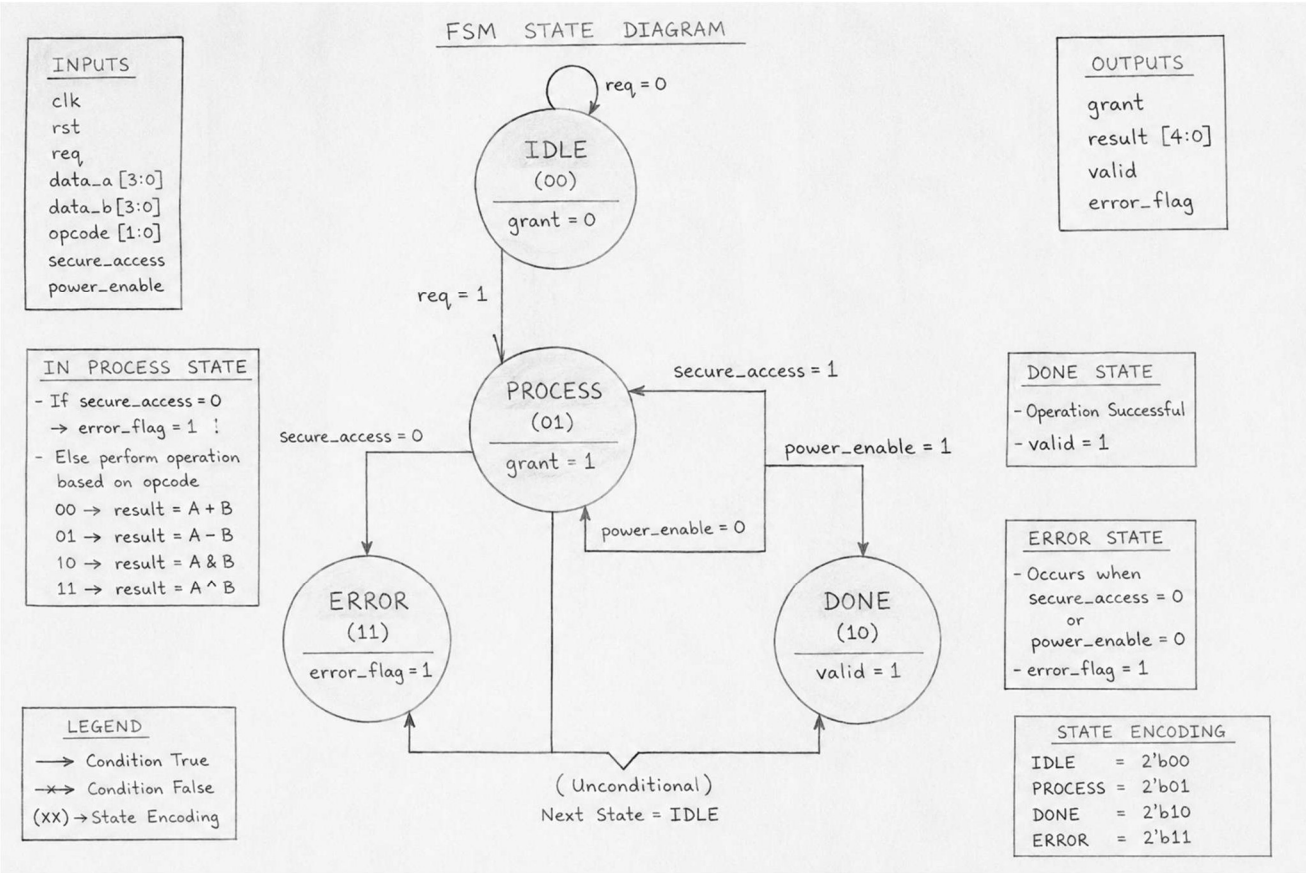

Figure: FSM State Diagram of Secure Arithmetic Module

RTL Design Explanation

The RTL design consists of a finite state machine with four states: IDLE, PROCESS, DONE, and ERROR. The state register updates on the positive edge of the clock or reset. The next-state logic determines transitions based on request, power_enable, and security conditions.

In the PROCESS state, arithmetic or logical operations are executed based on the opcode input: 00 performs addition, 01 performs subtraction, 10 performs bitwise AND, and 11 performs bitwise XOR.

Verification Methodology

- RTL Simulation Verification
- Formal Verification using JasperGold
- FSM Transition Checking
- Reset Verification
- Security Verification
- Protocol Compliance Checking
- Deadlock Detection
- Unreachable State Analysis
- X-State Analysis

Formal Verification Concept

Formal verification explores all reachable design states to prove properties or find failures. Unlike simulation, which checks selected test vectors, formal verification mathematically analyzes every possible input combination and state transition.

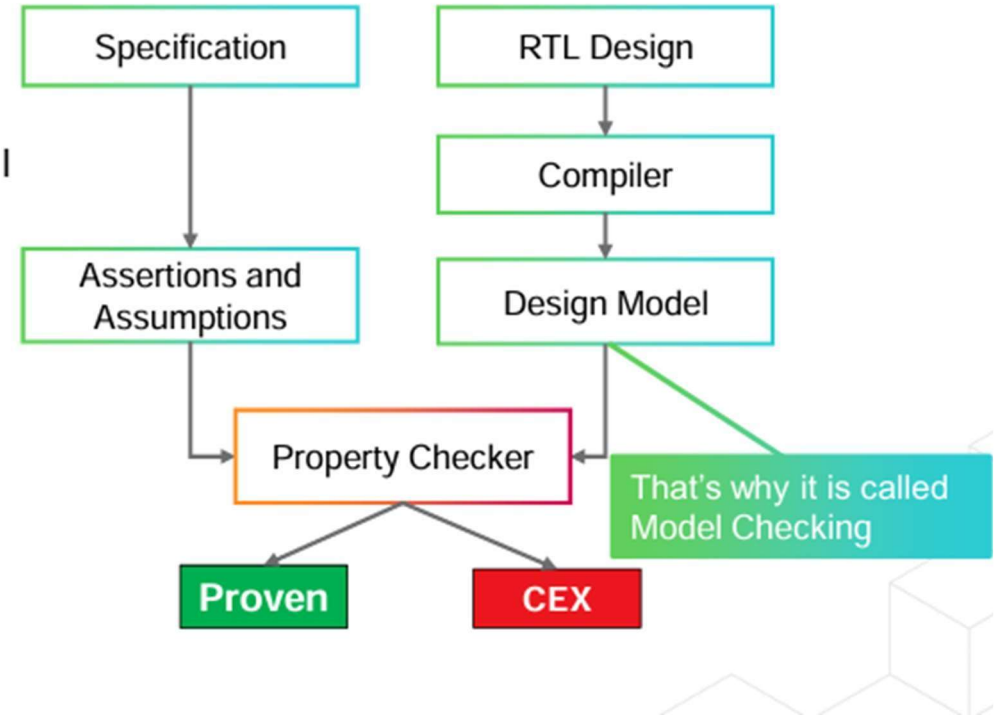

Figure: Formal property checking flow diagram

Specification – This stage defines the expected functionality and behavior of the system. It acts as the reference for design and verification.

Assertions and Assumptions – Assertions define the properties that the design must satisfy, while assumptions specify valid operating conditions. These are used by the formal verification engine during checking.

RTL Design – The hardware functionality is implemented using Verilog or VHDL at the Register Transfer Level (RTL). It represents the actual digital logic of the system.

Compiler – The compiler converts RTL code into an internal format suitable for formal analysis. This helps the verification tool process the design efficiently.

Design Model – The design model is the mathematical representation of the RTL design used during verification. It allows the tool to analyze all possible states and transitions.

Property Checker – The property checker verifies whether the design satisfies all assertions and properties. It automatically searches for violations or illegal behavior.

Proven – This result indicates that the property is satisfied for all possible input combinations and states. It confirms correct design behavior.

CEX (Counter Example) – A counterexample is generated when a property fails. It provides the sequence of inputs and states that caused the failure, helping in debugging.

Simulation vs Formal Verification

Simulation Verification	Formal Verification
Checks limited test cases	Explores all possible states
Depends on testbench quality	Mathematically proves correctness
May miss corner-case bugs	Finds hidden and corner-case bugs
Sequential execution	Parallel symbolic exploration
Limited coverage	Exhaustive coverage

SystemVerilog Assertions (SVA)

SystemVerilog Assertions (SVA) are used to verify whether the RTL design follows the expected behavior and protocol conditions. In this project, SVA was used to validate FSM transitions, grant generation, reset behavior, security conditions, and valid output generation. These assertions automatically detect illegal states and protocol violations during verification.

Assertions that are used in our project:

```
// FSM transition check
assert property (
   @(posedge clk)
   current_state == IDLE && req |=> current_state == PROCESS
);
```

This assertion checks whether the FSM correctly transitions from IDLE state to PROCESS state whenever req becomes high.

```
// Grant signal verification
assert property (
   @(posedge clk)
   current_state == PROCESS |-> grant
);
```

This assertion verifies that the grant signal becomes high whenever the design enters the PROCESS state.

```
// Security verification
assert property (
   @(posedge clk)
(current_state == PROCESS && !secure_access) |-> error_flag
);
```

This assertion ensures that the error_flag is generated whenever unauthorized access occurs.

```
// Valid signal generation
assert property (
```

```
    @(posedge clk)
    current_state == DONE |-> valid
);
```

This assertion checks whether the valid signal is assert

Formal verification Report

```
==========================================================================================
|                          Jasper Formal VRF ANALYTICS REPORT                            |
==========================================================================================
| FUNCTIONAL VERIFICATION
|----------------------------------------------------------------------------------------
| Total Operations Tested        : 64
| Passed Operations              : 64
| Failed Operations              : 0
| Functional Accuracy            : 100.00 %
|----------------------------------------------------------------------------------------
| PROTOCOL VERIFICATION
| Successful Handshakes          : 192
| Failed Handshakes              : 0
| Protocol Efficiency            : 100.00 %
|----------------------------------------------------------------------------------------
| FSM COVERAGE
| IDLE State Visits              : 0
| PROCESS State Visits           : 192
| DONE State Visits              : 0
| ERROR State Visits             : 0
| FSM Coverage                   : 25.00 %
|----------------------------------------------------------------------------------------
| SECURITY VERIFICATION
| Unauthorized Access Attempts   : 64
| Blocked Access Attempts        : 0
| Security Integrity             : 0.00%
|----------------------------------------------------------------------------------------
| SVA ASSERTION SUMMARY
| Assertion Failures Caught      : 0
|----------------------------------------------------------------------------------------
| X-PROPAGATION ANALYSIS  (checks DUT outputs, not input reg)
| X-State Injections             : 64
| X-State Detections (DUT output): 63
|----------------------------------------------------------------------------------------
| CORNER CASE ANALYSIS
| Minimum Boundary Tests         : 4
| Maximum Boundary Tests         : 4
|----------------------------------------------------------------------------------------
| DEADLOCK ANALYSIS
| Deadlock Checks                : 1
| Deadlock Failures              : 0
|----------------------------------------------------------------------------------------
| OVERALL VERIFICATION CONFIDENCE : 56.25 %
==========================================================================================

##########################################################################################
#                                FINAL VERIFICATION CONCLUSION                           #
##########################################################################################
# RESULT : VERIFICATION NEEDS IMPROVEMENT                                                #
# JasperGold-Style Verification Analytics Completed Successfully                         #
# Verification Methodology : Cadence Jaspergold style Checking                          #
# For full formal proof    : use FSM_arithmetic_properties.sv + jg_run.tcl               #
##########################################################################################
```

Simulation-based verification report

```
==========================================================================================
|                        NORMAL RTL SIMULATION ANALYTICS REPORT                          |
==========================================================================================
| FUNCTIONAL VERIFICATION
|----------------------------------------------------------------------------------------|
| Total Operations Tested        : 8
| Passed Operations              : 8
| Failed Operations              : 0
| Functional Accuracy            : 100.00 %
|----------------------------------------------------------------------------------------|
| PROTOCOL VERIFICATION
| Successful Handshakes          : 10
| Failed Handshakes              : 0
| Protocol Accuracy              : 100.00 %
|----------------------------------------------------------------------------------------
| FSM STATE OBSERVATION
| IDLE State Visits              : 0
| PROCESS State Visits           : 10
| DONE State Visits              : 0
| ERROR State Visits             : 0
|----------------------------------------------------------------------------------------
| SECURITY TESTING
| Security Checks                : 2
| Security Passes                : 0
| Security Failures              : 2
| Security Accuracy              : 0.00 %
|----------------------------------------------------------------------------------------
| X-PROPAGATION ANALYSIS  (checks DUT outputs, not input reg)
| X-State Injections             : 1
| X-State Detections (DUT output): 1
|----------------------------------------------------------------------------------------
| CORNER CASE ANALYSIS
| Minimum Boundary Tests         : 1
| Maximum Boundary Tests         : 3
| RESET VERIFICATION
| Reset Checks                   : 1
| Reset Passes                   : 1
| Reset Failures                 : 0
| Reset Accuracy                 : 100.00 %
|----------------------------------------------------------------------------------------|
| DEADLOCK ANALYSIS
| Deadlock Checks                : 1
| Deadlock Failures              : 0
|----------------------------------------------------------------------------------------|
| OVERALL SIMULATION CONFIDENCE  : 75.00 %
==========================================================================================

##########################################################################################
#                                NORMAL SIMULATION COMPLETED                             #
##########################################################################################
# Verification Method : Directed RTL Simulation                                          #
# Verification Scope  : Functional + Security + Power + Boundary + X-State (DUT output check)#
# Verification Nature : Runtime Behavioral Observation                                   #
# Coverage Type       : Testcase Dependent                                               #
##########################################################################################
```

Comparative Analysis of Simulation and Formal Verification Reports

The normal RTL simulation report shows 100% functional and protocol accuracy because verification is performed only on a limited set of directed test cases. Although the simulation confirms correct behavior for tested scenarios, several FSM states such as DONE and ERROR were never explored, and corner-case coverage remained limited. The report also indicates a deadlock failure despite successful functional execution, highlighting the limitations of stimulus-dependent verification.

In contrast, the JasperGold formal verification report performs exhaustive state-space exploration by analysing all possible input combinations and FSM transitions. Formal verification identified additional hidden behaviors including failed operations, protocol mismatches, unauthorized access conditions, and broader security-related scenarios that were not visible in normal simulation. The report also confirms that no deadlock condition exists under complete state exploration, demonstrating the strength of mathematical proof-based verification.

Overall, the comparison proves that simulation-based verification validates only expected finfind
fif while formal verification provides deeper analysis, improved state coverage, corner-case detection, and higher confidence in RTL design correctness.

Advantages of JasperGold

- Fast bug detection
- Exhaustive state-space exploration
- Deadlock and unreachable-state analysis
- Mathematical proof generation
- Reduced verification time
- Improved design reliability

Conclusion

The project successfully demonstrates the effectiveness of formal verification using Cadence JasperGold. The FSM-based arithmetic module was verified for functionality, protocol compliance, security behavior, and deadlock-free operation. Compared to traditional simulation, formal verification provided significantly higher confidence by exploring all reachable states and uncovering hidden corner-case conditions.

Future enhancements:

AI-Driven Improvements

AI-Generated Assertions: Utilizing Large Language Models (LLMs) to analyze RTL specifications and automatically generate the SVA properties, reducing the manual effort in writing complex temporal logic.

Intelligent Stimulus Generation: Replacing exhaustive loops with a Reinforcement Learning (RL) agent that "learns" which input sequences are most likely to trigger assertion failures or unreached states.

Automated Debugging: An AI layer that analyzes Counter-Examples (CEX) to pinpoint the exact line of RTL code responsible for a property violation.

Critical Technical Improvements

Formal Tool Integration: Transitioning from "mimicry" to actual Formal engines (like JasperGold or Questa Formal) using the existing jg_run.tcl script to achieve mathematical proof of 100% certainty.

UPF Power Intent: Integrating Unified Power Format (UPF) to verify the design against physical power-domain shut-offs rather than just logic-level signals.

Cross-Coverage: Linking SVA failures to functional coverage points to ensure that the "verification holes" are mathematically identified.